Pokéchess
UNITE

1

Project Description

POKÉMON UNITE

- **Pokémon UNITE** is a multiplayer online battle arena (MOBA) that matches **two teams** against each other.
- Each team consists of **five Pokémon** controlled by **different players**.
- The game map is split into **two halves**, with multiple **goal zones** on each side.
- Players score points for their team by **defeating "wild" Pokémon** and **dunking their held points** on an enemy goal.

POKÉMON UNITE

- A match lasts for **5 or 10 minutes**, depending on the game mode. The team with the **most points scored** on all enemy goals after the match time limit ends, wins the game.

POKÉCHESS UNITE

- In this project, you are tasked to make a **Pokémon UNITE-like** application, called **Pokéchess UNITE**.
- The **game specifications** are outlined in the provided **document**.
- To **simplify** this project, a **turn-based**, **chess-like mechanic** is used to replace the Pokémon movement in the original game.
- The original turn-based **Pokémon Battle mechanic** is also used to **replace** the fast-paced arena battle mechanic of the original game.

2

Game Mechanics

THE GAME MECHANICS

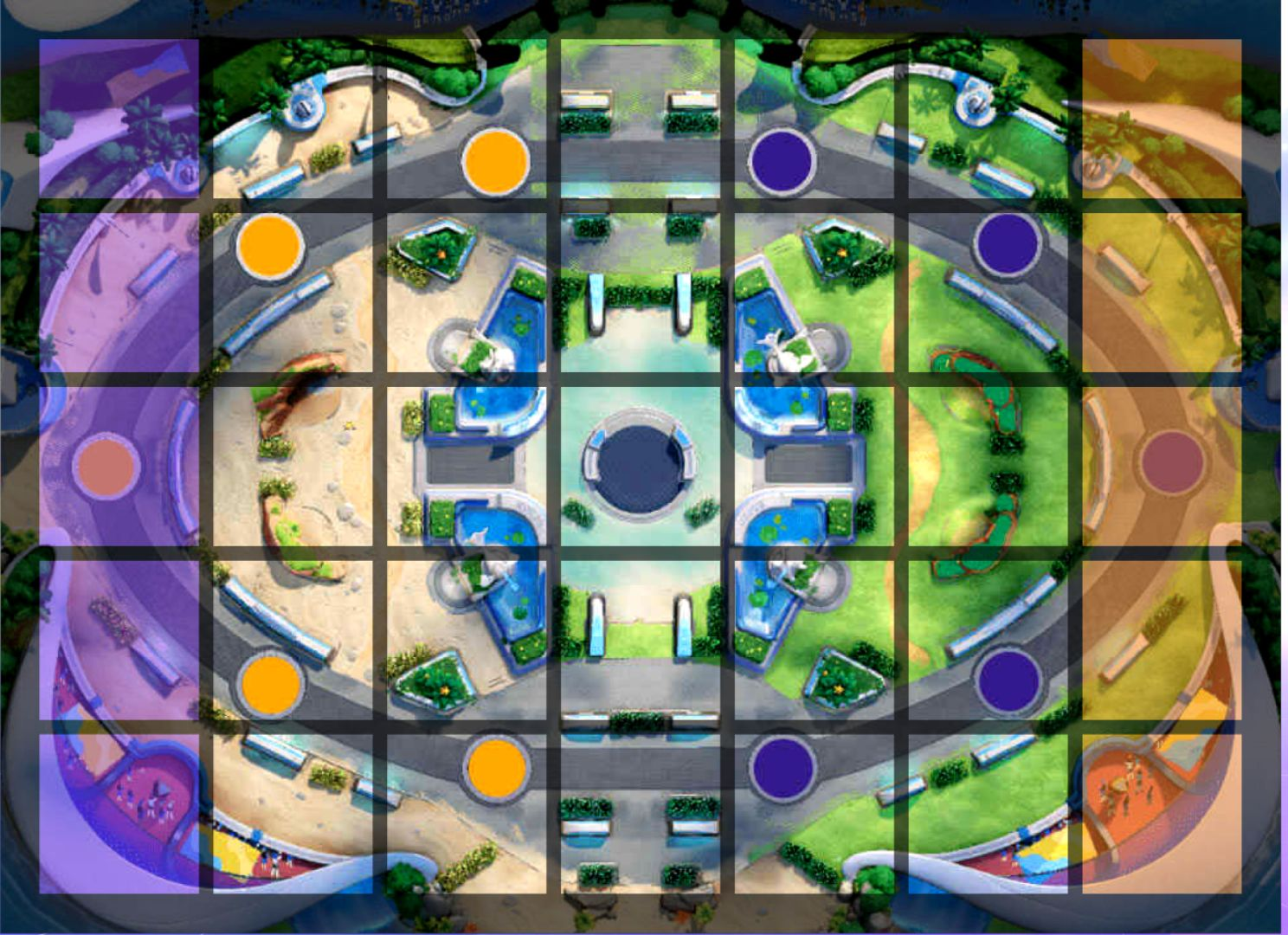
- Choosing a **Team**
- **Board** Mechanics
- **Battle** Mechanics
- **Scoring**

CHOOSING A **TEAM**

- Before the start of the game, the player creates a **team of 5 Pokémon** from the given **Pokémon Roster**.
- The **computer's team** of Pokémon is **generated randomly**.
- The player and computer can only have **at most 2 Pokémon** of the same **Battle Type**.
  - This means that a team cannot have 3 or more Attackers, 3 or more Speedsters, etc.

THE **BOARD** MECHANICS

- The match begins after the player has decided the composition of their team.
- A **5x7 Game Board** is shown at the start of the match. This is the **Board View**.

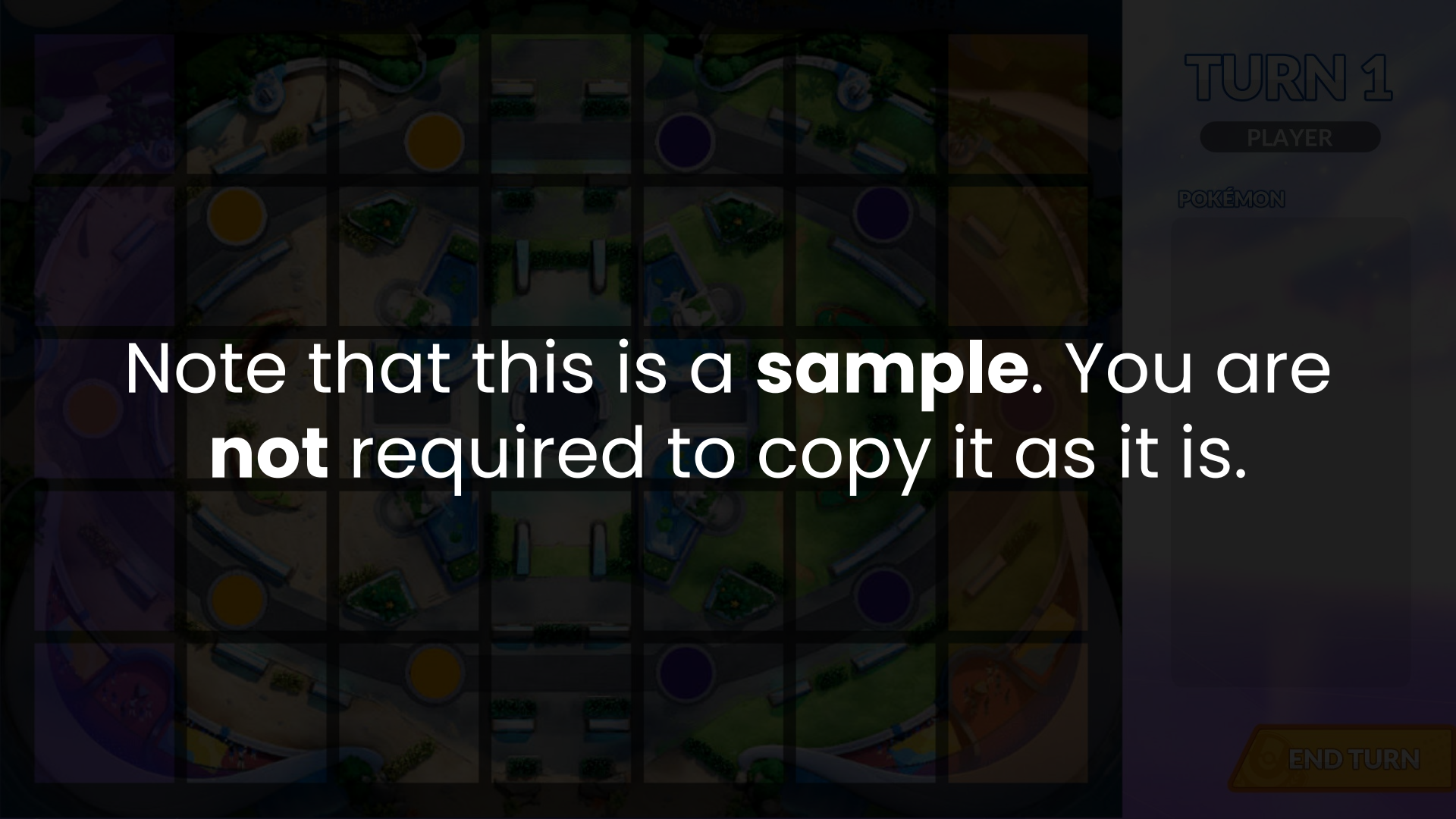


TURN 1

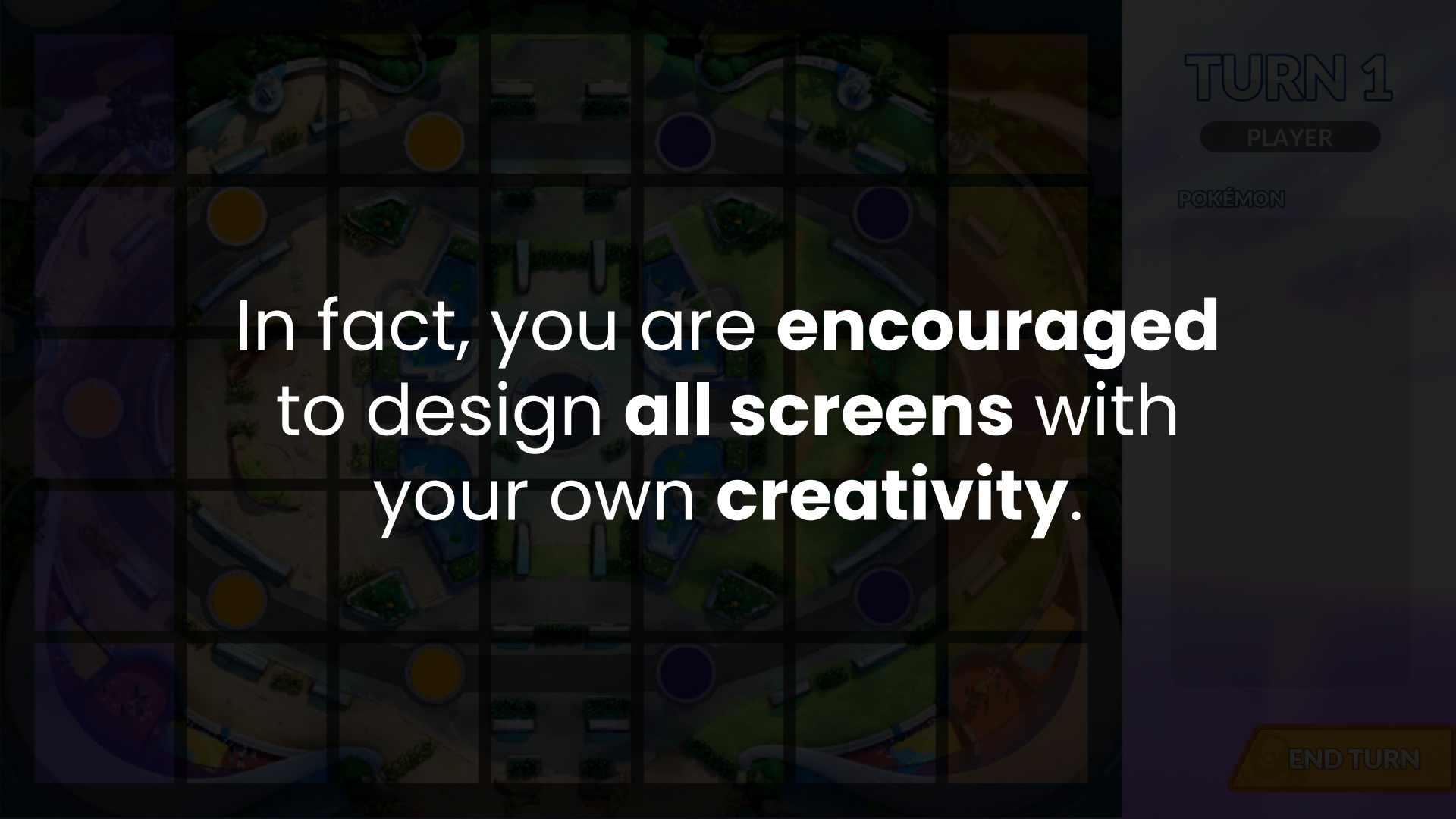
PLAYER

POKÉMON

END TURN



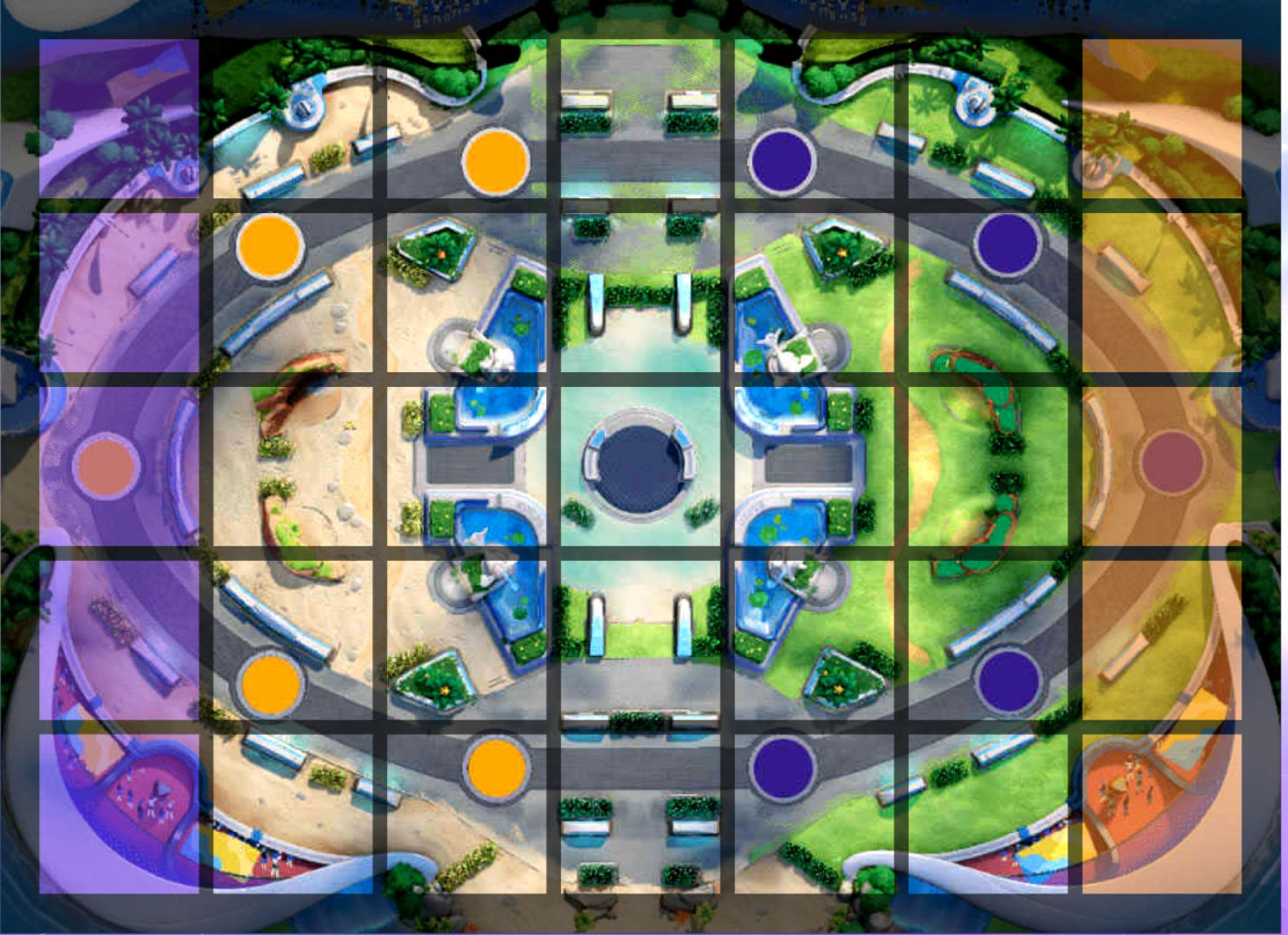
Note that this is a **sample**. You are **not** required to copy it as it is.



In fact, you are **encouraged** to design **all screens** with your own **creativity**.

THE **BOARD** MECHANICS

- Each team is on **one side** of the **shorter end** of the board, with the player's team on the left.



TURN 1

PLAYER

POKÉMON

END TURN

PLAYER'S TEAM
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
ENEMY'S TEAM
END TURN

BOARD MECHANICS

- The **column** in which the team is located at the start of the game is their **Home Zone**.

PLAYER'S HOME ZONE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
ENEMY'S HOME ZONE
END TURN

BOARD MECHANICS

- The **column** in which the team is located at the start of the game is their **Home Zone**.
- The **tile** in which a Pokémon is located at the start of the game is their **Home Tile**.

TURN 1
PLAYER
POKÉMON
END TURN

SYLVEON'S HOME TILE
TURN 1
PLAYER
POKÉMON
END TURN

BLASTOISE'S HOME TILE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
GENGAR'S HOME TILE
END TURN

TURN 1
PLAYER
POKÉMON
ELDEGOSS' HOME TILE
END TURN

TURN 1
PLAYER
POKÉMON
END TURN
GRENINJA'S HOME TILE

BOARD MECHANICS

- The **column** in which the team is located at the start of the game is their **Home Zone**.
- The **tile** in which a Pokémon is located at the start of the game is their **Home Tile**.
- The **team's Goal Zone** is also the **enemy's Home Zone**.

PLAYER'S HOME ZONE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
PLAYER'S GOAL ZONE
END TURN

TURN 1
PLAYER
POKÉMON
ENEMY'S HOME ZONE
END TURN

ENEMY'S GOAL ZONE
TURN 1
PLAYER
POKÉMON
END TURN

BOARD MECHANICS

- Note that **enemy Pokémon** are **not allowed** to step on the **player's Home Zone**.

TURN 1
PLAYER
POKÉMON
END TURN

SYLVEON'S HOME TILE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

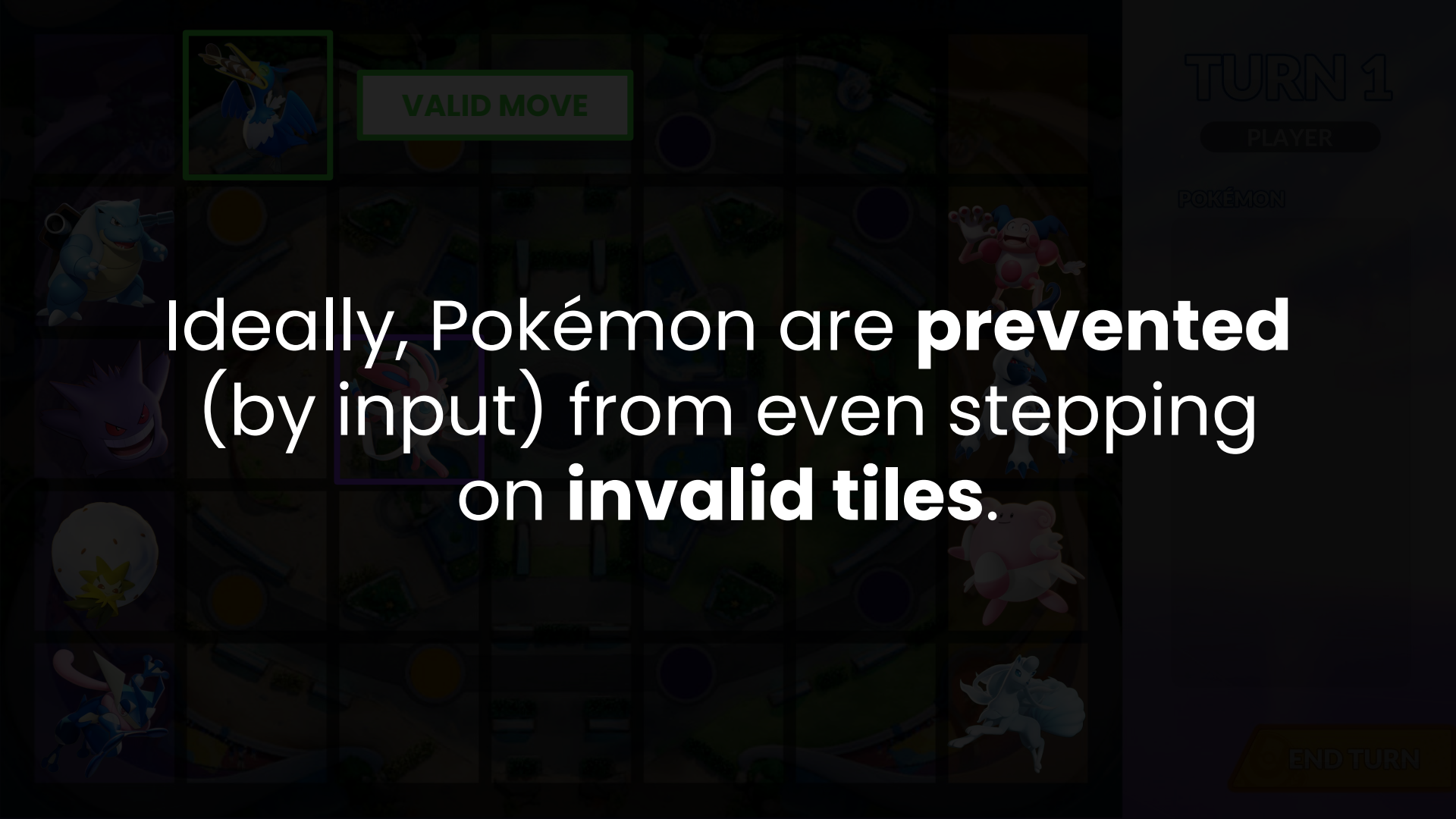
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

INVALID MOVE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

VALID MOVE
TURN 1
PLAYER
POKÉMON
END TURN



Ideally, Pokémon are **prevented** (by input) from even stepping on **invalid tiles**.

BOARD MECHANICS

- Note that **enemy Pokémon** are **not allowed** to step on the **player's Home Zone**.
- **Ally Pokémon** are only allowed to step at their **own Home Tile** at the Home Zone.

TURN 1
PLAYER
POKÉMON
END TURN

PLAYER'S HOME ZONE
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

INVALID MOVE
TURN 1
PLAYER
POKÉMON
END TURN

BOARD MECHANICS

- Note that **enemy Pokémon** are **not allowed** to step on the **player's Home Zone**.
- **Ally Pokémon** are only allowed to step at their **own Home Tile** at the Home Zone.
- A Pokémon's **Home Tile** is also considered as a **Valid Tile** for that Pokémon.

BOARD MECHANICS

- Note that **enemy Pokémon** are **not allowed** to step on the **player's Home Zone**.
- **Ally Pokémon** are only allowed to step at their **own Home Tile** at the Home Zone.
- A Pokémon's **Home Tile** is also considered as a **Valid Tile** for that Pokémon.
- Pokémon stepping at their Home Tile will **fully recover** their HP at the **end of each turn**, given that they have not died during the turn.

BOARD MECHANICS

- The team who makes the **first move** is decided **randomly**.
- On **each turn**, the player and computer can only use, **at most**, 3 Pokémon from their team.
  - See **Board Commands**.
- A Pokémon's **movement** depends on its **Battle Type**.

BOARD MECHANICS

- Similar to chess, a Pokémon can attempt to **kill an enemy Pokémon** by moving to the **same tile** it occupies.
- However, unlike chess, **killing is not automatic**. When a Pokémon attempts to kill an enemy Pokémon, a **Battle ensues** and the game switches to the **Battle Screen**.

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

ATTACK DEFEND HEAL RUN

BOARD MECHANICS

- The **Battle ends** when a **Pokemon's HP** turns **zero**.
- **After the battle**, the game switches back to the **Board View**.
- The **winning Pokémon** remains on the tile, while the **losing Pokémon** dies and is removed from the board.
  - A **dead Pokémon** cannot be used until it is **revived**.
  - A dead Pokémon **automatically revives** after a **number of turns**, depending on its Battle Type's **Revival Rate**.
  - A revived Pokemon appears at its **Home Tile** with **full Health**.

BOARD COMMANDS

- A **player** can issue **3 Board Commands** per turn.
- A Board Command is **consumed** whenever the player **uses a Pokémon to**:
  - Move
  - Initiate Battle
  - Dunk
  - Heal*
  - Activate its Unite Move*

* IF APPLICABLE

BOARD COMMANDS | MOVE

- Pokémon can be commanded to MOVE in the following directions:

[□]
FORWARD

Moving a Pokémon Forward moves it closer to the Goal Zone. When on the Player's side, this is a step towards the East

[□]
BACKWARD

Moving a Pokémon Backward moves it closer to the Home Zone. When on the Player's side, this is a step towards the West

[□]
UP

Moving a Pokémon Up moves it closer to the top of the board. This is a step towards the North

[□]
DOWN

Moving a Pokémon Down moves it closer to the bottom of the board. This is a step towards the South

[□ | □]
D. FORWARD

Pokémon can move Diagonally Forward [□|□]. When on the Player's side, this is a step towards NE or SE

[□ | □]
D. BACKWARD

Pokémon can move Diagonally Backward [□|□]. When on the Player's side, this is a step towards NW or SW

Let’s try moving **Gengar**.
He has a **Speed** of 3 Tiles.

TURN 1
PLAYER
POKÉMON
END TURN

VALID TILES FOR FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

VALID FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

VALID FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

INVALID FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN
VALID TILES FOR UP [□] MOVEMENT

TURN 1
PLAYER
POKÉMON
END TURN
VALID UP [□] MOVEMENT

TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN
VALID TILES FOR D.FORWARD [□ | □] MOVEMENT

TURN 1
PLAYER
POKÉMON
END TURN
VALID D.FORWARD [□] MOVEMENT

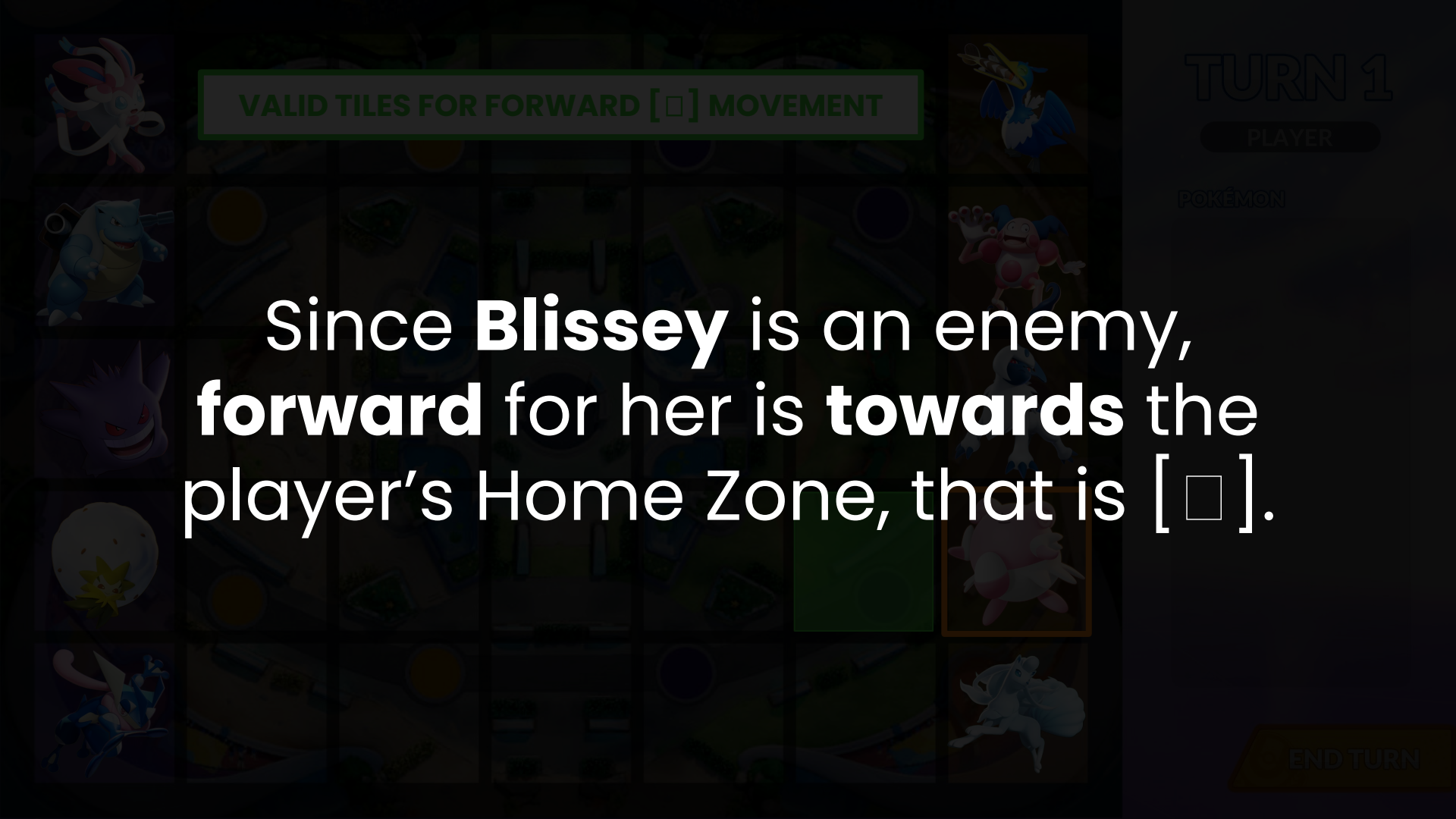
TURN 1
PLAYER
POKÉMON
END TURN
VALID D.FORWARD [□] MOVEMENT

TURN 1
PLAYER
POKÉMON
END TURN
INVALID D.FORWARD [□] MOVEMENT

Let’s try moving **Blissey**.
She has a **Speed** of 1 Tile.

TURN 1
PLAYER
POKÉMON
END TURN

VALID TILES FOR FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN



Since **Blissey** is an enemy, **forward** for her is **towards** the player's Home Zone, that is [□].

VALID TILES FOR FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

VALID FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMANDS | MOVE

- Pokémon are **NOT allowed** to **jump over** enemy Pokémon in the way of their **Maximum Tile Distance** (MTD).
  - MTD is the same as Speed. It just specifies all tiles a Pokémon's Speed can cover.

TURN 1
PLAYER
POKÉMON
END TURN

VALID TILES FOR FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

BUT IF AN ENEMY WAS IN THE WAY...
TURN 1
PLAYER
POKÉMON
END TURN

VALID TILES FOR FORWARD [□] MOVEMENT
TURN 1
PLAYER
POKÉMON
END TURN

INVALID TILE DESPITE BEING PART OF MTD
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMANDS | INITIATE BATTLE

- INITIATE BATTLE occurs when a Pokemon is commanded to **step on a tile** that an **enemy currently occupies**.
  - The INITIATE BATTLE command **leads to** the Battle View or **Battle Mechanic**.

TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMAND | MOVE [□]
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMAND | MOVE [□]
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMAND | INITIATE BATTLE
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMAND | INITIATE BATTLE
TURN 1
PLAYER
POKÉMON
END TURN

ATTACK DEFEND HEAL RUN

BOARD COMMANDS | DUNK

- A DUNK command **can only be issued** when the **following conditions** are met:

1. The Pokémon being commanded is **stepping** on one of the tiles **beside the Goal Zone**.
2. The Pokémon is **carrying at least 1 point**.

TURN 1
PLAYER
POKÉMON
10
0
1
5
11
END TURN

VALID DUNK COMMAND
10
0
1
5
11
TURN 1
PLAYER
POKÉMON
END TURN

CAN'T DUNK: 0 POINTS
10
0
1
5
11
TURN 1
PLAYER
POKÉMON
END TURN

CAN'T DUNK: OUT OF RANGE
10
0
1
5
11
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMANDS | DUNK

- If a player issues the DUNK command while an **enemy Defender** is at the **Goal Zone**, the points will not be added to the score yet.
- Instead, a **Battle** would automatically start between the **dunking Pokémon** and the **enemy Defender**.
- If the **ally Pokémon wins**, and **no other enemy Defender** remains at the Goal Zone, the dunked points are **finally added** to the team's score.

TURN 1
PLAYER
POKÉMON
5
END TURN

ATTACKER
POKÉMON
SUPPORTER
SPEEDSTER
5
SUPPORTER
ATTACKER
END TURN

DUNKING W/O DEFENDER
5
TURN 1
PLAYER
POKÉMON
END TURN

DUNKING W/O DEFENDER
PLAYER'S TEAM +5 POINTS
0
TURN 1
PLAYER
POKÉMON
END TURN

TURN 1
PLAYER
POKÉMON
5
END TURN

TURN 1
PLAYER
POKÉMON
DEFENDER
5
END TURN

DUNKING WITH DEFENDER
5
TURN 1
PLAYER
POKÉMON
END TURN

DUNKING WITH DEFENDER
ELDEGOSS VS SLOWBRO
5
TURN 1
PLAYER
POKÉMON
END TURN

ATTACK DEFEND HEAL RUN

SLOWBRO
ELDEGOSS WINS
ELDEGOSS
ATTACK
DEFEND
HEAL
RUN

DUNKING WITH DEFENDER

ELDEGOSS VS SLOWBRO

ELDEGOSS WINS

TURN 1

PLAYER

POKÉMON

END TURN

DUNKING WITH DEFENDER

ELDEGOSS VS SLOWBRO

ELDEGOSS WINS

PLAYER'S TEAM +5 POINTS

TURN 1

PLAYER

POKÉMON

END TURN

BOARD COMMANDS | DUNK

- However, if **another enemy Defender** is still at the Goal Zone, the dunking Pokémon **immediately battles** that Defender too, given that it wins the fight with the previous Defender.
- Note that **neither battles** are **counted as commands**, since they automatically occur as a result of the DUNK command.
- After a **successful dunk**, the ally Pokémon's **carried points** become zero, but it **stays at the same tile**, unless commanded to MOVE.

BOARD COMMANDS | HEAL

- A HEAL command **can only be issued** when the **following conditions** are met:
  1. The Pokémon being commanded has a **Battle Type** of **Supporter**.
  2. The Pokémon is **attempting to heal** an ally Pokémon **directly beside it**.
- Supporters **heal 20%** of the targeted Pokémon's **HP**.

TURN 1
PLAYER
POKÉMON
SUPPORTER
END TURN

HEAL COMMAND TILE RANGE
TURN 1
PLAYER
POKÉMON
END TURN

BOARD COMMANDS | HEAL

- Supporters can only heal **one ally** Pokémon **at a time**.
- However, they are allowed to heal **the same ally repeatedly**, given that the player has **enough** Board Commands to do so.
- Supporters can also **target an ally** with **full health**.
  - Doing so has **no effect** on the targeted ally, but still **consumes** a Board Command.

THE **BATTLE** MECHANICS

- Whenever a **Battle occurs**, the player is taken to the **Battle View**.
- Battles in Pokechéss UNITE follow a **simplified** version of the **original Pokémon Battle** mechanic.

THE **BATTLE** MECHANICS

The player can **issue** the following **Battle Commands** to a Pokémon:

- Attack
- Defend
- Heal*
- Run

* IF APPLICABLE

THE **BATTLE** MECHANICS

- Per turn, the **player** and **computer** selects **1 Battle Command** for their Pokémon to perform.
- After their selection, the commands are **executed** in the following **order of priority**:

DEFEND ➡ ATTACK ➡ HEAL ➡ RUN

BATTLE COMMANDS

- The **Battle Commands** are **defined** below:

ATTACK	This action **inflicts** **damage** to the enemy Pokémon.
DEFEND	This action **reduces** the total damage a Pokémon receives from an attack by **20%**.
HEAL	This action is only available to **Supporters**. It lets them **heal 20%** of their max HP.
RUN	This action lets the Pokémon **run away** from battle, returning it to its **Home Tile.** It has a **40% chance** to succeed.

ABOUT SCORING

- The **main goal** of the game is to **dunk Points** in front of the **Goal Zone**.
- When a successful dunk command is issued by an ally Pokémon, all the **points it carries** are added to the **team's Score**.
- When the **match ends**, the team with the **higher score wins**.

ABOUT SCORING

- In case of a **Tie** where both team's final scores end up the same, the team who **achieved the score first** wins the match.

NOTES ON **SCORING**

- At the **start of the match**, all Pokémon have **1 point**.
- Whenever a Pokémon **successfully defeats** an opposing Pokémon, **1 point** is added to the winning Pokémon's **carried points**.
- If the Pokémon **loses to an enemy Pokémon**, its carried points are **divided in half**, and then **rounded down**.
- The **max points** a Pokémon can carry at a time is **50**.
  - A Pokémon carrying 50 points **can still battle** enemy Pokémon, it's just that they **gain no further points** whenever they win.

3

Characters

THE CHARACTERS

Pokéchess UNITE uses the same **5 Battle Types** from Pokémon UNITE. These are:

- **Attacker**
- **Speedster**
- **All-Rounder**
- **Defender**
- **Supporter**

THE CHARACTERS

- Since the game does not completely follow Pokémon UNITE mechanics, **changes** have been made to **each Battle Type**.
  - These are described in the next slide.

POKÉMON **STATS** PER **BATTLE TYPE**

	HEALTH	ATTACK	DEFENSE	SPEED	HP REGEN	REVIVAL RATE
ATTACKER	75	40%	15%	2 Tiles	5%	2 Turns
SPEEDSTER	50	40%	5%	3 Tiles	5%	2 Turns
ALL-ROUNDER	75	30%	15%	2 Tiles	10%	3 Turns
DEFENDER	100	20%	25%	1 Tiles	5%	4 Turns
SUPPORTER	100	20%	5%	1 Tile	15%	3 Turns

4

Bonus Features

5

Deliverables

MCO1 DELIVERABLES

- For **MCO1A**:
  - You are required to submit a **UML Diagram** (saved as a PDF) demonstrating your design for this phase of the machine project.
  - **Proper use** of **UML symbols** and **relationships** must be followed. Inheritance and Abstraction are not yet required.
  - The UML must also **match** the submitted **implementation** for your MCO1B.

MCO1 DELIVERABLES

- For **MCO1B**:
  - You are required to submit a **ZIP archive** of your Java Project, along with **any external libraries** used, via Canvas. Note that external libraries must be approved by your teacher first.
  - **UI is required**, but **not necessarily GUI**. Groups may choose to present their project via TUI or GUI. The same UI criteria will be applied regardless of which UI style is used.
  - Additionally, Battle is still optional in this phase. You can **simulate** the **result of a battle** either by random or choosing who always wins. Then for the 2nd Phase, **replace** it with an actual **Battle Module** that satisfies the requirements stated in the specs.

MCO1 DELIVERABLES

- For **MCO1B**:
  - A **demo video** must also be included with the **ZIP archive** during the submission of this project.

MCO2 DELIVERABLES

- For **MCO2A**:
  - You are required to submit the **updated UML Diagram** (saved as a PDF) demonstrating your design for this phase of the machine project. Make sure your design has been updated with **Inheritance** or **Abstraction** concepts, **whenever necessary**.
  - **Proper use** of **UML symbols** and **relationships** must be followed.
  - The UML must also **match** the **implementation** of MCO2B.
  - You are **not required** to include **GUI classes** in your UML. However, if you decide to include **any GUI class**, it must be **complete**.

MCO2 DELIVERABLES

- For **MCO2B**:
  - You are required to submit a **ZIP archive** containing the **updated version** of the files you sent in MCO1B, **except** for the demo video.
  - A **live demo** will be held during Finals week.
  - **GUI is now required** during this phase of the project. Implementing parts of the program using TUI will merit **deductions** (unless explicitly stated to do so).

6

Appendices

Pokéchess
UNITE